

Belise Swartwood

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EDUCATION

Massachusetts Institute of Technology, Cambridge, MA

May 2028

Bachelor of Science in Mathematics and Finance

Relevant Coursework: Laboratory in Corporate Finance, Laboratory in Investments, Corporate Financial Accounting, Taxes & Business Strategy, Economic Analysis for Business Decisions, Mergers Acquisitions & Private Equity, Probability, Statistics

SKILLS

Technologies: Python, R, MATLAB, Excel, Powerpoint

Technical: Financial modeling: DCF, LBO, M&A, Financial statement analysis

EXPERIENCE

Massachusetts General Hospital

Feb 2025 - Dec 2025

ML Neuroscience Researcher - Boston, Massachusetts

- Developed predictive continuous LASSO regression model reaching 0.71 R^2 value to identify top 8 factors driving clinically significant depression treatment response
- Validated clinical significance of predictions using classification model that achieved an AUC of 0.84
- Processed and annotated 2 fMRI datasets to evaluate brain connectivity network patterns
- Presented research at Gerontological Society of America conference 2025
- First author of research manuscript for submission to scientific journal

Growing Islands: Rebuilding Beaches and Protecting Coastlines

Oct 2024 - Feb 2025

MIT Undergraduate Researcher - Cambridge, Massachusetts

- Designed and tested physical prototypes to minimize sediment displacement by 20% and optimize wave disruption by 15%
- Conducted controlled lab experiments and numerical simulations to analyze sediment movement
- Trained ML model on coastal imagery to classify and predict sediment transport patterns

Hawaii State House Representative

Jun - Aug 2024

Legislative Intern - Honolulu, Hawaii

- Increased social media engagement by 50% and audience reach by 30% through targeted campaign strategy
- Analyzed voter outreach data to optimize marketing platform usage
- Assisted in community engagement through canvassing and event coordination

PROJECTS

Twitter LBO Model Case Study

Feb 2026

- Built a comprehensive LBO model to evaluate the \$44B acquisition of Twitter, incorporating multi-tranche debt structures, revolving credit facilities, and interest components to determine IRR and cash-on-cash returns across various exit scenarios.
- Conducted sensitivity analysis on key value drivers, including DAU (Daily Active User) growth and ad-revenue margins, while assessing the impact of significant cost-cutting measures on the company's ability to service high-yield debt obligations.

Merger Arbitrage Investment Strategy Research

Nov - Dec 2025

- Quantitative Backtesting & Strategy Implementation: Developed and backtested the "ArbiTIM" merger arbitrage strategy using a sample of 285 all-cash M&A deals (\$500M+ value) from 2015-2024, achieving results characterized by low market correlation and reduced volatility.
- Data-Driven Risk Management: Leveraged R-Studio and Bloomberg Terminal to analyze risk-adjusted returns and deal-failure probabilities, identifying key regulatory and execution risks to optimize a portfolio designed for long-term institutional investors.

ACTIVITIES

- MIT Varsity Swimming & Diving
 - NCAA DIII Finalist, 2025 Scholar All-American
- MIT Chamber Music Society; MIT Emerson/Harris Scholar, Violinist
- President, MIT Hawaii Club